MILITARY SPECIFICATION

WARHEAD, GUIDED MISSILE, INERT, XM10E3, PARTS FOR

1 SCOPE

1.1 This specification covers quality assurance provisions and special requirements not covered by the drawings for the fabrication of the parts for one type of warhead designated as Warhead, Guided Missile, Inert, XM10E3.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein:

SPECIFICATIONS

MILITARY

MIL-D-1000 — Drawing, Engineering and Associated List.

MIL-A-2550 — Ammunition and Special Weapons, General Specification for.

MIL-I-45607 — Inspection Equipment, Supply and Maintenance of.

STANDARDS

MILITARY

MIL-STD-105 — Sampling Procedures and Tables for Inspection by Attributes.

MIL-STD-109 — Inspection Terms and Definitions.

MIL-STD-1167 — Ammunition Data Cards.

MIL-STD-1235 — Single and Multi-level Continuous Sampling Procedures and Tables for Inspection by Attributes.

DRAWINGS

ORDNANCE CORPS

8800265 — Cushion.

8830879 — Crate, Packing, Ammunition, Dual, for Warhead, Guided Missile, HE, XM5E5, XM9E3 or Warhead, Guided Missile, Inert, XM10E3.

FSC 1336

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8837892 — Crate, Packing, Ammunition, Single, for Warhead, Guided Missile, HE, XM-5, XM9, or Warhead, Guided Missile, Inert, XM10.

8862440 — Warhead, XM10E3, Inert Parts Assembly.

PUBLICATIONS

ORDNANCE CORPS

IEL–8862440 — Index of Inspection Equipment Lists.

(Copies of specifications, standards, drawings and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Material. Materials and parts shall be in accordance with applicable drawings and specifications.

3.2 Assemblies. The warhead assemblies shall comply with all requirements specified on Drawing (Dwg.) 8862440 and with all requirements specified in applicable specifications.

3.3 Workmanship.

3.3.1 *Molded parts.* The molded parts shall be free of porosity, warpage in excess of drawing dimensions, checks, cracks, chipped edges, blisters, scratches and other defects that would affect their serviceability.

3.3.2 *Foreign matter.* All parts shall be free of chips, dirt, grease, excess flash, rust, silicone products and other foreign matter. The cleaning method used shall not be injurious to any part nor shall the parts be contaminated by the cleaning agent. Surface coatings shall be continuous except for a few light scratches not exposing base material. All marking shall be neat and sharply defined.

3.4 First article inspection. This specification makes provisions for first article inspection. Requirements for the submission of first article samples by the contractor shall be as specified in the contract.

4. QUALITY ASSURANCE PROVISIONS

4.1 General quality assurance provisions. Unless otherwise specified in the contract or purchase order, the supplier is responhible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements. Reference shall be made to Standard MIL–STD–109 in order to define terms used herein. The provisions of Specification MIL–A–2550 shall apply.

4.1.1 *Submission of product.* At the time the completed lot of product is submitted to the Government for acceptance, the contractor shall supply the following information accompanied by a certificate which attests that the information provided is correct and applicable to the product being submitted:

(a) A statement that the lot complies with all of the quality assurance provisions specified within this specification.

(b) Number of units of product inspected.

(c) Results obtained, by defect code, for all inspections performed.

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(d) Drawing, specification number and date, together with an identification and date of changes.

(e) Certificates of conformance on all material purchased by the contractor when such material is controlled by Government or commercial specifications referenced in any of the contractual documents.

(f) Number of items in the lot.

(g) Date submitted.

The certificate shall be signed by a responsible agent of the certifying organization. The initial certificate submitted shall be substantiated by evidence of the agent's authority to bind his principal. Substantiation of the agent's authority will not be required with subsequent certificates unless, during the course of the contract, this authority is vested in another agent of the certifying organization.

4.1.2 *Government verification.* Using the contractor's written quality assurance procedure, this detailed specification, the applicable drawings and other contractual documents as a guide, the Government inspector shall verify at unscheduled intervals, all quality assurance operations performed by the contractor. Verification will be performed to the extent necessary to assure compliance with the contractual requirements. Severity of Government inspection of individual characteristics will be directly related to the seriousness of the classification assigned. In no instance will a characteristic classified "critical" be accepted solely on the basis of the contractor's records.

4.2 First Article Inspection.

4.2.1 *Submission.* When items to this specification are not produced concurrently with XM5E5 or XM9E3 warheads having an approved initial production sample, the contractor shall submit a first article sample as designated by the contracting officer for evaluation in accordance with the provisions of 4.2.2. The first article sample shall consist of 2 assemblies and 4 complete sets of parts (i.e. 4 each of every component and every sub-assembly) which have been produced by the contractor or furnished by a supplier and which have been manufactured using the same production processes, procedures and equipment which will be used in fulfilling the contract. All parts and materials, including packaging and packing, shall be obtained from the same source of supply as will be used in regular production. Prior to submission, the contractor shall inspect the sample to the degree necessary to assure that it conforms to the requirements of the contract and submit a record of this inspection with the sample, including statements of findings/certificates of conformance for materials, (see 4.1.1.(e)). A sample containing known defects will not be submitted unless specifically authorized, by the Contracting Officer. A first article sample, or portion whereof, as directed by the contracting officer, shall also be submitted whenever there is a lapse in production for a period in excess of 90 days or whenever a change occurs in manufacturing process, material used, drawing or specification such as to significantly affect product uniformity as determined by the Government.

4.2.2 *Inspections to be performed.* Assemblies and components will be submitted by the Government to any or all of the examinations or tests specified in 4.3.2 and 4.3.3 of this specification and any or all requirements of the applicable drawings.

4.3 Inspection provisions.

4.3.1 *Lot formation.* The term "lot" as used throughout this specification refers to an inspection lot, which is defined as an essentially homogeneous collection of units of product from which a representative sample is drawn and inspected to determine conformance of the lot with applicable requirements. The sample selected shall represent only that quantity of units from which the

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sample was drawn and shall not be construed to represent any prior or subsequent quantities presented for inspection. Homogeneity shall be considered to exist provided the lot has been produced by one manufacturer, in one unchanged process, in accordance with the same drawing, specification, or revision thereof. Changes to either the process, specification, or drawing not affecting safety, performance, interchangeability, or storage, as determined by the Government inspector, shall not be deemed to alter the homogeneity of the lot. Inspection lots shall comply with MIL–STD–105 and shall be numbered in accordance with MIL–STD–1168.

Unless otherwise approved by the contracting officer, the inspection lot size of major assemblies or end items deliverable under the contract shall be not less than the smallest weekly estimate of quantities contractually scheduled for production during the contract period nor more than the largest quantity contractually scheduled for delivery during any month of the contract period. Inspection lots for components or subassemblies, other than the items of delivery, shall be homogeneous and of a size convenient to the contractor.

4.3.2 *Examination* Sampling plans and procedures for the following classifications of defects shall be in accordance with Standard MIL–STD–105 except that inspection for critical defects when listed, shall be 100 percent. Continuous sampling plans in accordance with Standard MIL–STD–1235 may be used if approved by the procuring activity. Also, at the option of the procuring activity, AQLs and sampling plans may be applied to the individual characteristics listed using an AQL of 0.40 percent for each major defect and an AQL of 0.65 percent for each minor defect. Once agreement has been reached as to which method will be used (group or individual) that method will be used for all of the CDs listed in this specification. Equipment necessary for the performance of the inspections listed shall be in accordance with 4.3.4.

4.3.2.1 Insert, forward (see dwg. 8800268 covering a detail of dwg. 8862440).

Categories	*Defects*	*Method of Inspection*	*Code No. (see 6.2)*
Critical: None defined.			
Major:	AQL 0.25 percent		
101.	Pitch diameter of internal thread	Gage	01001
102.	Major diameter of external thread, minimum (min.)	Gage	01002
Minor:	AQL 1.00 percent		
201.	Operation missing	Visual	01003
202.	Evidence of poor workmanship (see 3.3.2)	Visual	01004

4.3.2.2 Insert, mounting (see dwg. 8861590 covering a detail of dwg. 8862440).

Categories	*Defects*	*Method of Inspection*	*Code No.*
Critical: None defined.			
Major:	AQL 0.40 percent		
101.	Pitch diameter of threads	Gage	02001
102.	Thickness of flange	Gage	02002
103.	Diameter over knurl	Gage	02003
104.	Knurl incomplete or not sharp	Visual	02004
Minor:	AQL 1.50 percent		
201.	Minor diameter of thread, maximum (max.)	Gage	02005
202.	Diameter of flange, min.	Gage	02006
203.	Operation missing	Visual	02007
204.	Evidence of poor workmanship (see 3.3.2)	Visual	02008

4.3.2.3 Insert, aft (see dwg. 8800274 covering a detail of dwg. 8862440).

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Categories	Defects	Method of Inspection	Code No.
Critical: None defined.			
Major:	AQL 0.25 percent		
	101. Diameter of cavity	Gage	03001
	102. Depth of cavity, min.	Gage	03002
Minor:	AQL 1.50 percent		
	201. Chamfer missing	Visual	03003
	202. Knurl missing	Visual	03004
	203. Evidence of poor workmanship (see 3.3.2)	Visual	03005

4.3.2.4 Insert, liner (see dwg. 8800275 covering a detail of dwg. 8862440).

Categories	Defects	Method of Inspection	Code No.
Critical: None defined			
Major:	AQL 0.25 percent		
	101. Pitch diameter of thread	Gage	04001
	102. Diameter over knurl	Gage	04002
	103. Knurl incomplete or not sharp	Visual	04003
Minor:	AQL 1.00 percent		
	201. Minor diameter of thread max.	Gage	04004
	202. Evidence of poor workmanship (see 3.3.2)	Visual	04005

4.3.2.5 Body, inner, forward (see dwg. 8861587 covering a detail of dwg. 8862440).

Categories	Defects	Method of Inspection	Code No.
Critical: None defined			
Major:	AQL 0.25 percent		
	101. Sidewall thickness, min.	Gage	05001
	102. Molding defects (see 3.3.1)	Visual	05002
Minor:	AQL 0.40 percent		
	201. Foreign matter (see 3.3.2)	Visual	05003

4.3.2.6 Plug, closing (see dwg. 8800284 covering a detail of dwg. 8862440).

Categories	Defects	Method of Inspection	Code No.
Critical: None defined			
Major:	AQL 0.25 percent		
	101. Thickness, min.	Gage	06001
	102. Component damaged or not flat	Visual	06002
Minor:	AQL 0.40 percent		
	201. Foreign matter (see 3.3.2)	Visual	06003

4.3.2.7 Body, inner, aft (see dwg. 8861588 covering a detail of dwg. 8862440).

Categories	Defects	Method of Inspection	Code No.
Critical: None defined			
Major:	AQL 0.40 percent		
	101. Depth to bottom of insert, min.	Gage	07001
	102. Sidewall thickness, min.	Gage	07002
	103. Molding defects (see 3.3.1)	Visual	07003
Minor:	AQL 0.40 percent		
	201. Foreign matter (see 3.3.2)	Visual	07004

4.3.2.8 Liner, safety and arming (see dwg. 8861586 covering a detail of dwg. 8862440).

Categories	Defects	Method of Inspection	Code No.
Critical: None defined			
Major:	AQL 0.65 percent		
	101. Location of liner insert	Gage	08001

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Categories	Defects	Method of Inspection	Code No.
	102. Spring pin projects into safety and arming cavity	Visual	08002
	103. Insert above flush	Visual	08003
	104. Spring pin missing	Visual	08004
	105. Molding defect (see 3.3.1)	Visual	08005
Minor:	AQL 0.40 percent		
	201. Foreign matter (see 3.3.2)	Visual	08006

4.3.2.9 Assembly, warhead, inert parts, prior to inserting inserts and assembling fragments (see dwg. 8862440).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major:	AQL 0.65 percent		
	101. Diameter of forward insert hole, max.	Gage	09001
	102. Depth of forward insert hole, max.	Gage	09002
	103. Safety and arming liner not seated on flange or loose	Visual-manual	09003
	104. Foreign matter (see 3.3.2)	Visual	09004
	105. Diameter of mounting insert hole, max.	Gage	09006
	106. Depth of mounting insert hole, max.	Gage	09007
Minor:	AQL 0.40 percent		
	201. Evidence of poor workmanship (see 3.3.2)	Visual	09005

4.3.2.10 Assembly, warhead, inert parts, prior to painting (see dwg. 8862440).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major:	AQL 0.65 percent		
	101. Location of mounting inserts	Gage	10001
	102. The plane formed by the centerlines of the four mounting holes not perpendicular to centerline of warhead or plane formed by these centerlines not flat within tolerance specified	Gage	10002
	103. Total length, max.	Gage	10003
	104. Diameter across mounting inserts, max.	Gage	10004
	105. Forward insert above flush	Visual	10005
	106. Mounting flange cracked or damaged	Visual	10006
	107. Mounting insert pin missing or loose	Visual-manual	10007
Minor:	AQL 1.50 percent		
	201. Voids not filled	Visual	10008
	202. Outer body improperly applied	Visual	10009
	203. Evidence of poor workmanship (see 3.3.2)	Visual	10010

4.3.2.11 Asembly, warhead inert parts (see dwg. 8862440).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major:	AQL 0.40 percent		
	101. Weight marked on the assembly incorrect or weight of assembly out limits	Balance	11001
	102. Interior coating (see 6.4)	Visual	11002
	103. Paint or foreign matter in threaded holes, aft insert safety and arming liner or closing plug surface (see 3.3.2)	Visual	11003

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Categories	Defects	Method of inspection	Code No.
Minor:	AQL 1.50 percent		
201.	Closing plug insecurely taped	Visual-Manual	11004
202.	Exterior coating incomplete, inadequate or improper	Visual	11005
203.	Marking incomplete or incorrect	Visual	11006
204.	Foreign matter (see 3.3.2)	Visual	11007

4.3.2.12 Shipping container, prior to sealing (see dwg. 8830879 or dwg. 8837892, as applicable).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major:	AQL 0.40 percent		
101.	Cushioning pad on base missing	Visual	12001
102.	Warhead not seated properly	Visual	12002
103.	Hold down strap missing or insecure	Visual-manual	12003
Minor:	AQL 1.00 percent		
201.	Foreign matter in container	Visual	12004

4.3.2.13 Shipping container (see dwg. 8830879 or dwg. 8837892).

Categories	Defects	Method of inspection	Code No.
Critical:	None defined		
Major:	AQL 0.40 percent		
101.	Strapping missing, broken or loose	Visual-manual	13001
102.	Board broken, split or cracked	Visual	13002
103.	Skid missing or damaged	Visual	13003
104.	Contents exposed	Visual	13004
Minor:	AQL 1.50 percent		
201.	Strapping improperly assembled	Visual	13005
202.	Nails improperly set or missing	Visual	13006
203.	Marking incomplete or incorrect	Visual	13007

4.3.3 *Testing.*

4.3.3.1 First article sample (see 4.2.1). The samples submitted as the first article quantity shall be tested as specified below after having been examined for all other drawing requirements. Rejection and resubmission shall be in accordance with 4.2.3.

4.3.3.1.1 Presence of mounting insert pin (see dwg. 8862440 and 4.4.1) Code No. 11010.

4.3.3.1.2 Weight of assembly (see dwg. 8862440 and 4.4.2) Code No. 11009.

4.3.3.2 *Production testing.*

4.3.3.2.1 Presence of mounting insert pin (see dwg. 8862440) Major defect Code No. 11010. This determination shall be performed 100 percent as specified in 4.4.1 using the applicable equipment of 4.3.4. Any assembly from which one or more pins are missing shall be classed defective and removed from the lot pending repair and resubmission of test.

4.3.3.2.2 *Weight of assembly.* (See dwg. 8862440.) Major defect Code No. 11009. This determination shall be performed 100 percent as specified in 4.4.2. Any unit which fails to comply with the drawing requirement shall be classed defective and removed from the lot.

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4.3.3.2.3 Molded components (see dwgs. 8861586, 8861587, and 8861588). At the start of production from a new mold or cavity (each of which shall be identified) or after adjustment or repair to a mold or cavity, consecutively produced units from each mold or cavity shall be dimensionally examined until 4 consecutive units meet all applicable drawing requirements. After adjustment or repair, those dimensions affected by the adjustment or repair shall be examined. During production, the molding procedures shall be monitored for uniformity of molding practices. Suspected deviations in mold product shall be examined to determine if corrective action or mold repair is required.

4.3.4 *Inspection equipment.* Index of inspection equipment Number IEL–8862440 identifies the inspection equipment required to perform the examinations and tests prescribed in this section. The contractor shall design inspection equipment in accordance with the instructions in 6.3.

4.3.4.1 *Government rights to documentation.* Inspection equipment drawings and lists provided and revised in accordance with the requirements of the IEL may be used by DOD activities for design, procurement, manufacture, testing, evaluation, production and receiving inspection, overhaul, shipping, storage, identification of stock, ordering and storage of replacement parts, inspection items at overhaul, general maintenance of equipment, construction, survey and whenever inspection equipment drawings are needed.

4.3.4.2 *Supply and maintenance.* Supply and maintenance of the equipment listed in the ET shall be in accordance with Specification MIL–I–45607.

4.3.4.3 *Government use of contractor's inspection equipment.* The contractor shall make available to the Government all inspection equipment necessary for determining conformance with contract requirements. Personnel for operating the equipment and verification of its accuracy, shall be supplied by the contractor for the performance of examination or test by the Government.

4.4 Test methods and procedure.

4.4.1 *Presence of mounting insert pin.* The cavity of each mounting insert shall be examined using the gage specified in 4.3.4. If the shoulder of the gage contacts one or more inserts of a unit, that unit shall be classed defective.

4.4.2 *Weight of assembly.* The assembly shall be weighed using equipment accurate to within 0.01 pound. Any asembly, the weight of which is outside the range specified, shall be classed defective.

5. PREPARATION FOR DELIVERY

5.1 Packing and marking.

5.1.1 *Level C.* The assemblies shall be packed and marked in accordance with dwg. 8830879 or dwg. 8837892, as applicable.

5.2 Data cards. Data card information shall be as specified in Specification MIL–A–2550.

6. NOTES

6.1 Ordering data. Procurement documents shall specify the following:

(a) Title, number and date of this specification.

(b) Data cards. Data cards shall be prepared for each lot in accordance with the information specified in Standard MIL–STD–1167.

6.2 Inspection code numbers. The five digit code numbers assigned to the inspections herein are to facilitate future data collection and analysis by the Government.

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6.3 Inspection equipment. The contractor shall design inspection equipment as required by the referenced inspection Equipment Lists (EL) in accordance with the instructions of paragraphs 6.3.1 through 6.3.5.

6.3.1 *Inspection equipment lists.* Inspection equipment lists indicate the availability of inspection equipment designs by showing in the "number" column of the list of inspection equipment (Form SMUPA 1010) the numbers of drawings or Federal Stock numbers of existing equipment designs, or codes as indicated in paragraph 6.3.2. Design action required of the contractor is described in paragraphs 6.3.3 and 6.3.4. The contractor will be required to prepare detailed drawings in accordance with 6.3.4 for all the equipment coded as "Contractor Design" in the number column. These contractor designs must be approved by the Government prior to fabrication or procuring of the equipment. Designs shall be submitted for approval as specified in 6.3.5.

6.3.2 *Inspection equipment list codes.* The inspection equipment as defined in 6.3.3 and 6.3.4 will be designated in the EL by the following codes:

CD — Contractor design on controlled contractor format and/or commercial equipment.

MU — Army design, mandatory for use.

6.3.3 *Army designs.* Army designs are reflected on detailed drawings which completely depict all the information necessary for the fabrication of the item of inspection equipment. The contractor need provide no design when an Army design is listed for an item of inspection equipment. Army designs fall into two basic classifications; mandatory (designated "MU") and nonmandatory (indicated by drawing or Federal Stock Number). When an inspection equipment list references mandatory Army designs, the contractor shall comply with, and use these designs accordingly. The contractor may, however, in connection with nonmandatory designs, and with the approval of the Government, design alternate inspection equipment or use comparable commercial equipment to facilitate his operations. Such contractor prepared designs or commercial equipment selections must be approved by the Government prior to fabrication or procuring of the equipment. Designs shall be submitted for approval as specified in 6.3.5.

6.3.4 *Contractor designs.* Contractor designs are designs of inspection equipment for which the Government has assigned design responsibility to the contractor. Contractor designs shall be supported by detailed drawings which depict all information necessary to completely fabricate, calibrate and operate an item of inspection equipment. This requires that the necessary views, dimensions, materials, finish, notes, operating and calibration instructions be properly depicted in accordance with approved practices to the extent that further calculation or clarification will not be required. Unless otherwise specified, contractor designs may be developed on the format the contractor normally employs in his equipment design procedure provided such format reflects the detail and information specified above, subject to the following controls: All submitted contractor designs shall conform to MIL-D-1000, Category E, Form 2. Legibility and reproducibility shall permit conventional making of clearly understandable, high contrast reproductions. Contractors shall submit one copy of final designs as a rolled set. Designs shall be submitted for approval as specified in 6.3.5.

6.3.5 *Submission of contractor designs.* All submitted designs shall contain a reference to the applicable five digit code number in section 4 of this specification. Unless otherwise specified on the EL, all designs of equipment for inspection of defects classified as critical and major shall be submitted for approval to the Commanding Officer, Picatinny Arsenal, ATTN: SMUPA-ND1. All other

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designs of inspection equipment shall be approved by the inspection element of the agency administering the contract; submission shall be as directed by the contracting officer. Partial submission of inspection equipment designs is permissible and encouraged. However, the Arsenal completion date for design review will be based on the date of the final submission of designs. Picatinny Arsenal design review will be accomplished normally within one month after receipt.

6.4 Interior surface. Paint on the interior surface of the assembly shall not be considered objectionable.

6.5 Shipping containers. The procuring activity should notify the parts contractor of the possible availability of used wood shipping containers.

Notice. When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission or manufacture, use, or sell any patented invention that may in any way be related thereto.

Custodian:

Army—Ordnance Corps

Preparing activity:

Army—Ordnance Corps

Project No. 1336–A–118

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